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### Guidewire for cannula placement

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#### Abstract

A guidewire for backloading and inserting a percutaneous pump affixed to a cannula includes a proximal section made of a first material, with a first diameter, a rounded proximal end, and a distal end. The guidewire also includes a distal section made of a second material, with a second diameter which is greater than the first diameter, a distal end, and a proximal end abutting the distal end of the proximal section. The first material of the proximal section is selected to be softer than a material of the percutaneous pump to reduce damage to the pump during backloading. The distal section of the guidewire is configured to be stiffer than the proximal section to insert the percutaneous pump in a desired location without damaging the guidewire.

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**Background/Summary**

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application is a continuation of U.S. patent application Ser. No. 17/232,425, filed on Apr. 16, 2021, now U.S. Pat. No. 11,752,308, which is a continuation of U.S. patent application Ser. No. 16/381,942, filed on Apr. 11, 2019, now U.S. Pat. No. 11,007,350, which is a continuation of U.S. patent application Ser. No. 15/790,107, filed Oct. 23, 2017, now U.S. Pat. No. 10,300,249, which is a continuation of U.S. patent application Ser. No. 14/862,090, filed on Sep. 22, 2015, now U.S. Pat. No. 9,821,146, the disclosures of all of which are hereby incorporated by reference in their entirety.

**BACKGROUND OF THE INVENTION**

(1) A blood pump, such as a percutaneous intracardiac blood pump assembly, is introduced in the heart to deliver blood from the heart into an artery, such as the pulmonary artery. When deployed in the heart, a blood pump assembly pulls blood from the left ventricle of the heart and expels blood into the aorta, or pulls blood from the right ventricle and expels blood into the pulmonary artery, through a cannula. Blood pump assemblies are introduced surgically or percutaneously during a cardiac procedure through the vascular system. In one common approach, pump assemblies are inserted by a catheterization procedure through the femoral artery using a guidewire.

(2) An introducer is inserted into the femoral artery through an arteriotomy to create an insertion path. A placement guidewire is then advanced distal end first through the sheath into the artery. Once the guidewire has been inserted into the artery, the pump assembly, including a pump and a

cannula, is backloaded onto the proximal end of the guidewire and pushed along the guidewire and into the patient. The pump assembly may then be used in conjunction with a catheter.

(3) As defined herein, backloading involves inserting the proximal end of the guidewire, which remains outside the patient, into the distal end of the catheter and then advancing the catheter distally over the wire. Backloading the pump assembly allows the guidewire to remain in position within the patient while different size catheters or sheaths are inserted and removed during the course of a procedure. But the cannula of the pump assembly can have a tortuous shape, or the cannula can be stiff. In these cases, the guidewire may not be stiff enough to allow backloading of the pump through to the pulmonary valve, and as a result the cannula of the pump may displace the guidewire out of the pulmonary valve. For example, for systems delivering blood from the inferior vena cava to an opening in the pulmonary artery, the cannula of the pump may be a 3D shape with two “S” turns in different planes. This can make backloading and insertion of the pump assembly into a patient particularly difficult.

#### SUMMARY OF THE INVENTION

(4) Systems, methods, and devices for an improved guidewire for cannula placement are presented. The improved guidewire facilitates insertion of the guidewire through a heart pump without damaging the pump. This improved guidewire is particularly helpful for pumps having complex or tortuous geometries, such as the IMPELLA RP pump or any other pump adapted for use in the right ventricle (e.g., between the inferior vena cava and the pulmonary artery).

(5) The improved guidewire disclosed herein can be inserted into the arterial system of a patient through an arteriotomy. The guidewire includes a first distal section, an intermediate pump delivery section, and a backloading proximal section. The first distal section, is inserted first into the arterial system of the patient. The first distal section is flexible and has a rounded end which allows a physician to insert the guidewire with minimal trauma to the patient. For example, during insertion the first distal section of the guidewire may come into contact with the walls of the patient's artery, or with a lumen placed inside the patient's artery. Thus, use of a material with low friction and high flexibility for the first distal section, such as a plastic or polymer, can reduce trauma and discomfort to the patient.

(6) The intermediate pump delivery section, is connected to the first distal section. While the first distal section of the guidewire facilitates insertion in the patient, the intermediate pump delivery section facilitates backloading of a pump assembly onto the guidewire. The pump assembly includes a pump affixed to a cannula. During backloading, the proximal end of the guidewire which remains outside the patient must be inserted into the pump and through the cannula, before the pump and cannula can be pushed along the guidewire until reaching the desired site. Depending on the application, the cannula may take different shapes. For example, in the case of some pumps for the right ventricle (e.g., the IMPELLA RP pump), the pump is located at one end of a cannula with a complex three-dimensional shape, including two “S” bends in different planes. Having an intermediate pump delivery section of the guidewire which is stiffer than the first distal section facilitates backloading of the stiff cannula and minimizes the risk of the stiff cannula displacing the guidewire. The pump delivery section can have a larger diameter than the first distal section and can be made of a stiffer material.

(7) The intermediate section of the guidewire is connected to the backloading proximal section. While the intermediate section, also referred to as intermediate pump delivery section, facilitates guiding the pump assembly into position without displacing the guidewire, the backloading proximal section minimizes damage to the pump during the backloading process. Due to its softness and flexibility relative to the intermediate pump delivery section, the backloading proximal section can be inserted into the pump without damaging the pump. This is particularly helpful because the small size of the pump and the guidewire routinely complicate initiation of the backloading process. Inserting the guidewire into the pump through the small gap between the impeller blades and the housing may require several attempts.

- (8) In some implementations, the first distal section, the intermediate pump delivery section, and the backloading proximal section of the guidewire have different stiffnesses due to differences in materials, structures, geometries, or a combination thereof.
- (9) The guidewire disclosed herein provides a number of possible advantages. The guidewire is flexible enough at its proximal end to pass through the pump without unnecessary damage. At the same time, the guidewire is stiff enough at its distal end to guide the cannula into place during backloading without being displaced. This can help avoid multiple insertions of the guidewire into the patient and minimizes the risk of damage to the patient's arterial system.
- (10) In one aspect, a system for inserting a percutaneous pump includes a percutaneous pump, a cannula and a guidewire. The cannula has a cannula diameter, a proximal inlet, and a distal outlet. The percutaneous pump is located at, and affixed to, the distal outlet of the pump. The guidewire includes a proximal section with a first stiffness and having a first diameter, and a distal section connected to the proximal section, the distal section with a second stiffness and a second diameter which is greater than the first diameter. The distal section is stiffer than the proximal section to insert the cannula and position it in a desired location without displacing the guidewire.
- (11) In certain implementations, the proximal section of the guidewire uses a first material which is softer than a material of the percutaneous pump to reduce damage to the percutaneous pump during backloading of the cannula onto the guidewire.
- (12) In certain implementations, the proximal section is made of a first material with the first stiffness and the distal section is made of a second material with the second stiffness.
- (13) In certain implementations, the proximal section is made of a first structure with the first stiffness and the distal section is made of a second structure with the second stiffness.
- (14) In certain implementations, the guidewire includes a distal tip which is connected to a distal end of the distal section.
- (15) In certain implementations, the cannula has a three-dimensional shape with a first "S" bend in a first plane, and a second "S" bend in a second plane, where the second plane is different from the first plane.
- (16) In certain implementations, the proximal section has a rounded proximal end made of a third material.
- (17) In certain implementations, a ratio of a diameter of the proximal section over a diameter of the distal section is 0.72.
- (18) In certain implementations, a stiffness of the proximal section is lower than a stiffness of the distal section.
- (19) In certain implementations, the proximal section extends over 40 to 75% of a total length of the guidewire.
- (20) In another aspect, a guidewire for backloading and inserting a percutaneous pump includes a proximal section and a distal section. The proximal section is made of a first material and has a first diameter. The proximal section includes a rounded proximal end and a distal end. The distal section is made of a second material and has a second diameter. The second diameter of the distal section is greater than the first diameter of the proximal section. A proximal end of the distal section abuts the distal end of the proximal section. The first material of the proximal section is selected to be softer than a material of the percutaneous pump to reduce damage to the pump housing or blades during backloading of the percutaneous pump onto the guidewire. The distal section is configured to be stiffer than the proximal section such that the percutaneous pump can be inserted and moved to a desired location without displacing the guidewire.
- (21) In some implementation, a stiffness of the second material is greater than a stiffness of the first material.
- (22) In some implementation, the proximal section is made of a first structure and has a first stiffness and the distal section is made of a second structure and has a second stiffness.
- (23) In some implementation, the proximal section is made of a first structure and has a first

stiffness, and the distal section is made of a second structure and has a second stiffness.

(24) In certain implementations, a stiffness of the proximal section is lower than a stiffness of the distal section.

(25) In certain implementations, the proximal section is coated and the distal section is not coated.

(26) In certain implementations, a ratio of a diameter of the proximal section over a diameter of the distal section is 0.72.

(27) In certain implementations, the proximal end extends over 40 to 75% of a total length of the guidewire

(28) In certain implementations, the proximal section includes a distal tip which is connected to a distal end of the distal section.

(29) In certain implementations, the rounded proximal end of the proximal section is made of a third material.

(30) In another aspect, a method for inserting a percutaneous pump includes inserting a guidewire into a cannula through a distal end of the cannula which supports the percutaneous pump, and pushing the guidewire through the percutaneous pump and cannula. The guidewire includes a proximal section with a first stiffness and a first diameter, and a distal section connected to the proximal section, the distal section with a second stiffness and a second diameter greater than the first diameter. The distal section is stiffer than the proximal section to insert the cannula in a desired location without displacing the guidewire.

(31) Variations and modifications will occur to those of skill in the art after reviewing this disclosure. The disclosed features may be implemented, in any combination and subcombination (including multiple dependent combinations and subcombinations), with one or more other features described herein. The various features described or illustrated above, including any components thereof, may be combined or integrated in other systems. Moreover, certain features may be omitted or not implemented.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

(1) The foregoing and other objects and advantages will be apparent upon consideration of the following detailed description, taken in conjunction with the accompanying drawings, in which like reference characters refer to like parts throughout, and in which:

(2) FIG. 1 shows an illustrative embodiment of a cannula assembly;

(3) FIG. 2 shows a lateral cross-section of a conventional guidewire;

(4) FIGS. 3 and 4 show illustrative damage to blades of a percutaneous pump, resulting from backloading the percutaneous pump on a conventional guidewire;

(5) FIG. 5 shows a lateral cross-section of a first illustrative embodiment of a guidewire;

(6) FIG. 6 shows a lateral cross-section of a second illustrative embodiment of a guidewire;

(7) FIG. 7 shows a lateral cross-section of a third illustrative embodiment of a guidewire;

(8) FIG. 8 shows a lateral cross-section of a fourth illustrative embodiment of a guidewire;

(9) FIGS. 9 and 10 show tables of data related to properties of the guidewire in FIG. 2 and a guidewire in any of FIGS. 2-8;

(10) FIG. 11 shows a percutaneous pump being backloaded on a guidewire according to one of the illustrative embodiments of FIGS. 5-8; and

(11) FIG. 12 shows an illustrative process for inserting a guidewire.

### DETAILED DESCRIPTION

(12) To provide an overall understanding of the systems, methods, and devices described herein, certain illustrative embodiments will be described. Although the embodiments and features described herein are specifically described for use in connection with a percutaneous blood pump

system for the right ventricle, it will be understood that all the components and other features outlined below may be combined with one another in any suitable manner and may be adapted and applied to blood pump systems for the left ventricle or other types of cardiac therapy and cardiac assist devices, including balloon pumps, cardiac assist devices implanted using a surgical incision, and the like.

(13) The systems, methods, and devices described herein provide a guidewire having a first proximal section and a second distal section such that a cannula can be inserted over a guidewire without displacing the guidewire and without damaging a pump coupled to the cannula. The proximal section of the guidewire is less stiff than the distal section of the guidewire due to the geometry or the material of the proximal section. For example, the proximal section of the guidewire may have a diameter which is smaller than a diameter of the distal section. In another example, the proximal section of the guidewire may be made of a material which is less stiff than a material of the distal section of the guidewire. In yet another example, the proximal section of the guidewire may have a structure which is less stiff than a structure of the distal section of the guidewire. The lower stiffness of the proximal section, relative to the stiffness of the distal section, allows physicians to more easily insert the guidewire into a percutaneous pump located at a distal end of a cannula. In particular, physicians may be able to insert the proximal section of the guidewire into the gap located between the housing and the blades of the percutaneous pump without damaging the blades, and without damaging or rupturing the housing.

(14) FIG. 1 shows an illustrative embodiment of a blood pump assembly **100**. The blood pump assembly **100** includes a pump **101**, a pump housing **103**, a proximal end **105**, a distal end **107**, a cannula **108**, an impeller **109**, an extension **102**, a catheter **112**, an inlet area **110**, an outlet area **106**, and blood exhaust apertures **117**. The catheter **112** is connected to the inlet area **110** of the cannula **108**. The inlet area **110** is located near the proximal end **105** of the cannula, and the outlet area **106** is located toward the distal end **107** of the cannula **108**. The inlet area **110** includes a pump housing **103** with a peripheral wall **111** extending about a rotation axis **113** of the impeller blades **115**, positioned radially outward of the inner surface with respect to the rotation axis **113** (not shown) of the impeller **109** (not shown). The impeller **109** is rotatably coupled to the pump **101** at the inlet area **110** adjacent to the blood exhaust apertures **117** formed in the wall **111** of the pump housing **103**. The pump housing **103** may be composed of a metal in accordance with implementations. The extension **102**, also referred to as a “pigtail,” is connected to the distal end **107** of the cannula **108** and assists with stabilizing and positioning the blood pump assembly **100** into the correct position in the heart. The pigtail **102** is configurable from a straight to a partially curved configuration. The pigtail **102** may be composed, at least in part of a flexible material, and may have dual stiffness.

(15) The cannula **108** has a shape which matches the anatomy of the right ventricle of a patient. In this exemplary embodiment, the cannula has a proximal end **105** arranged to be located near the patient's inferior vena cava, and a distal end **107** arranged to be located near the pulmonary artery. The cannula **108** includes a first segment **S1** extending from the inflow area to a point B between the inlet area **110** and the outlet area **106**. The cannula **108** also includes a second segment **S2** extending from a point C, which is between the inlet area **110** and the outlet area **106**, to the outlet area **106**. In some implementations B and C may be at the same location along cannula **108**. The first segment **S1** of the cannula forms an ‘S’ shape in a first plane. In some implementations, segment **S1** can have curvatures between 30 degrees and 180 degrees. The second segment **S2** of the cannula forms an ‘S’ shape in a second plane. In some implementations, segment **S2** can have curvatures between 30 degrees and 180 degrees 40°, 50°, 60°, 70°, 80°, 90°, 100°, 110°, 120°, 130°, 140°, 150°, 160°, or 170°). The second plane can be different from the first plane. In some implementations, the second plane is parallel or identical to the first plane.

(16) In some implementations, the blood pump assembly **100** is inserted percutaneously through the femoral artery and into the right ventricle. Alternatively, in some implementations the blood

pump assembly **100** may be inserted percutaneously through the femoral artery and into the left ventricle. When properly positioned, the blood pump assembly **100** delivers blood from the inlet area **110**, which sits inside the patient's left ventricle, through the cannula **108**, to the blood exhaust apertures **117** of the pump housing **103** positioned in the ascending aorta.

(17) FIG. 2 shows a lateral cross-section of a conventional guidewire **200**. The conventional guidewire includes a distal section **210**, a distal tip **212**, a coil wire **214**, a core wire **216**, a transition region **218**, a proximal section **220**, a proximal end **222**, and a transition section **218**. The coil wire **214** surrounds the core wire **216**. The core wire **216** has a diameter which decreases from the transition region **218** to the tip of the distal section **212**. The distal section **210** has a length  $L_2$ , which is between 25-50% of the total length of the core wire **216**. The core wire **216** also includes a proximal section **220**, extending between a proximal end **4222** and the transition section **218**. The proximal section **220** has a constant diameter. The proximal section **220** has a length between 75-50% of the total length of the core wire **216**. The distal section **210** of the core wire is more flexible than the proximal section of the guidewire **200**. This allows a physician initially placing the guidewire into a patient to minimize damage to the arterial system of the patient.

(18) FIGS. 3 and 4 show illustrative damage to a blade of a percutaneous pump. Exemplary blades **310** and **430** include portions **320** and **440** which were scratched or dented or both by attempting to insert a conventional guidewire between the pump housing (e.g., pump housing **103** in FIG. 1) and the blades (e.g., impeller blades **115** in FIG. 1). In this example, contact between the guidewire and a pump element results in the pump element being scratched or indented (as shown in FIGS. 2 and 3) rather than in the guidewire being scratched or indented. Although only scratches and indentations are shown, in some cases the pump housing can be punctured when inserting a conventional guidewire. This is particularly a concern when using a guidewire made of a material with a higher scratch hardness, indentation hardness or rebound hardness than a material used for the pump elements.

(19) As discussed above, when a guidewire (e.g., guidewire **200** in FIG. 2) is not stiff enough, backloading and insertion of a pump assembly (e.g., pump assembly **100** in FIG. 1) into a patient may be particularly difficult. Accordingly, some physicians may use a stiffer guidewire to backload a pump (e.g., pump **101** in FIG. 1) without displacing the guidewire **200** out of the pulmonary valve. However, for particular pumps, such as the IMPELLA RP pump which is used in combination with an "S" turn cannula (e.g., cannula **108** in FIG. 1), a stiffer guidewire may not be a viable solution. In the exemplary embodiment of the IMPELLA RP pump (e.g., pump **101** in FIG. 1), the pump is so small that the passage between the pump housing (e.g., pump housing **103** in FIG. 1) and the blades of the pump (e.g., impeller blades **115** in FIG. 1) is on the order of millimeters. Furthermore, due to their size and complexity the components of the IMPELLA RP pump such as the housing (e.g., housing **103** in FIG. 1) and the blades (e.g., impeller blades **115** in FIG. 1) are particularly expensive to manufacture or replace. Any undesired contact between the guidewire and the pump elements, instead of having the guidewire pass straight through the aforementioned gap, can result in damage. This is particularly true for a stiff guidewire, such as a guidewire made of a material with a higher scratch hardness, indentation hardness or rebound hardness than a material used for the pump elements. A guidewire with increased stiffness throughout, either via an increased diameter or via material properties, could damage the housing or blades of the pump even more than a conventional guidewire.

(20) FIG. 5 shows a lateral cross-section of a first illustrative embodiment of a guidewire **500**. The guidewire **500** includes a soft distal section **510**, a distal tip **512**, a coil wire **514**, a core wire **516**, a transition area **518**, an intermediate pump delivery section **530**, a transition area **532**, a backloading proximal section **540**, and a proximal tip **542**. The soft distal section **510** extends between the transition area **518** and the distal tip **512**. In use, the soft distal section **510** is inserted first into the patient. The soft distal section **510** includes the core wire **516** and the coil wire **518** coiled around the core wire **516**. The core wire **516** has a diameter which decreases from the transition region **518**



to the distal tip **512**. Instead of a single proximal section as in the guidewire **200**, the guidewire **500** includes two sections, the intermediate pump delivery section **530** and the backloading proximal section **540**. The intermediate pump delivery section **530** and the backloading proximal section **540** are coupled to the intermediate pump delivery section **530** by a transition area **532**. The intermediate pump delivery section **530** extends between the transition area **532** and the transition area **518**. The intermediate pump delivery section **530** is the portion of the guidewire **500** with the widest diameter. The intermediate pump delivery section **530** may have a constant diameter. In some implementations, the intermediate pump delivery section **530** is not the widest section of the guidewire.

(21) The backloading proximal section **540** extends between the proximal tip **542** and the transition section **532**. The backloading proximal section **540** has a constant diameter. In some implementations, the diameter of the proximal section **540** varies. The material and the structure of the backloading proximal section **540** and the intermediate pump delivery section **530** may be similar or the same. The backloading proximal section **540** is more flexible than the intermediate pump delivery section **530** because the backloading proximal section **540** has a diameter which is smaller than the diameter of the intermediate pump delivery section **530**. The smaller diameter of the backloading proximal section facilitates the introduction of the guidewire through the pump with low forces. In some implementations, the backloading proximal section **540** is more flexible than the intermediate pump delivery section **530** because the backloading proximal section **540** is formed of a material or a structure which is less stiff than the material or structure of the intermediate pump delivery section **530**.

(22) In certain implementations, the backloading proximal section **540** is more flexible than the intermediate pump delivery section **530** by having a material composition which is 100% a material which is more flexible than a material of the intermediate pump delivery section **530**. The intermediate pump delivery section **530** can have a material composition which is 100% a material stiffer than the material of the backloading proximal section **540**. The stiffer material of the intermediate pump delivery section **530** may consist of any number of materials including but not limited to polyurethane or resin impregnated fibers. The more flexible material may consist of any number of materials including but not limited to a silicone compound. In some implementations, the two different materials may have the same chemical composition, but different degrees of polymerization, crystallinity, or any other property.

(23) The material of the backloading proximal section **540** may be selected to reduce damage to the pump elements when there is undesired contact between the guidewire and any of the pump elements. In particular, the material of the backloading proximal section can be selected to have a lower scratch hardness, indentation hardness or rebound hardness than a material used for the pump elements.

(24) The proximal tip **542** of the backloading proximal section **540** is rounded and is made with or coated with a material different from the remainder of the backloading proximal section **540**. For example, a lubricating coating or lubricated material can be used for the proximal tip **542**. This allows a physician to insert the proximal tip **542** and the backloading proximal section **540** of the guidewire more easily into the pump and the cannula. In turn, these features can help reduce damage to the blades and housing of the percutaneous pump into which the guidewire is inserted. The proximal tip **542** can be attached to the backloading proximal section **540** by adhesive or solvent bonding, mechanical fastening, insert molding, or any other suitable joining mechanism or combinations thereof. Alternatively, the proximal tip **542** can be integral to the backloading proximal section.

(25) FIG. **6** shows a lateral cross-section of a second illustrative guidewire **600** according to certain embodiments. The guidewire **600** includes a soft distal section **610**, a distal tip **612**, a coil wire **614**, a core wire **616**, a transition area **618**, an intermediate pump delivery section **630**, a transition area **632**, a backloading proximal section **640**, and a proximal tip **642**. The guidewire **600** includes a

distal section **610** which extends between the transition area **618** and the distal tip **612** and which is inserted first into the patient. The soft distal section **610** includes a core wire **616** with a diameter which decreases from the transition region **638** to the distal tip **612**. The soft distal section **610** also includes a coil wire **614** which is coiled around the core wire **616**. The soft distal section **610** extends over a length **L2**, which can be 25-50% (e.g., 30%, 35%, 40%, 45%) of the total length of the guidewire **600**. Preferably, the length **L2** is between 25-35% of the total length of the guidewire **600**. Even more preferably, the length **L2** is 30% of the total length of the guidewire **600**. The guidewire **600** further comprises, instead of the proximal section **220** for example, two sections, the intermediate pump delivery section **630**, and the backloading proximal section **640** coupled to the intermediate pump delivery section **630** by a transition area **632**. The intermediate pump delivery section **630** extends between the transition area **632** and the transition area **618**. The intermediate pump delivery section **630** can have a constant diameter **D3**, which may be between 0.02" and 0.03", with a preferred value of 0.025". The intermediate pump delivery section **630** has a length **L3** which is between 30% and 60% (e.g., 35%, 40%, 45%, 50%, 55%) of the length of the guidewire. In particular, **D3** can be larger than any diameter of the backloading proximal section **640**. The backloading proximal section **640** has a constant diameter **D4**, which may be between 0.017" and 0.019" with a preferred value of 0.018". The backloading proximal section **640** can have a length **L4** which is 40-75% (e.g., 45%, 50%, 55%, 60%, 65%, 75%) of the length of the guidewire. For example, for a guidewire with a total length of 260 mm, the backloading proximal section **640** has a length of at least 100 mm.

(26) At least one advantage of the diameter **D4** of the backloading proximal section **640** being smaller than the diameter **D3** of the intermediate pump delivery section **630** is increased flexibility of the backloading proximal section **640**. Another advantage is a reduction in the overall weight of the guidewire. This reduction in weight is achieved because the backloading proximal section **640** is made lighter, and that backloading proximal section accounts for a substantial portion of the overall length of the guidewire **600** (e.g., 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, or more). A lighter guidewire is easier for physicians to insert into a patient and may also cause less damage when inserted into a pump, such as the pump **101** of the blood pump assembly **100**.

(27) The backloading proximal section **640** may end with the proximal tip **652**. The proximal tip **642** may be rounded and may be made with or coated with a different material than a material of the backloading proximal section **640**. For example, a lubricating coating or lubricated material can be used for the proximal tip **652**. This allows a physician to insert the proximal tip and the backloading proximal section of the guidewire more easily into the pump and the cannula. In turn, these features can help reduce damage to the blades and housing of the percutaneous pump the guidewire is being inserted into.

(28) FIG. 7 shows a lateral cross-section of a third illustrative guidewire **700** according to certain embodiments. The guidewire **700** includes a soft distal section **710**, a distal tip **712**, a coil wire **714**, a core wire **716**, a transition area **718**, an intermediate pump delivery section **730**, a transition area **732**, a backloading proximal section **740**, and a proximal tip **742**. The guidewire distal section **710** extends between the transition area **718** and the distal tip **712**, and is inserted first into the patient. The soft distal section **710** can include a core wire **716** with a diameter which decreases from the transition region to the distal tip **712**. The soft distal section **710** includes the coil wire **714** which is coiled around the core wire **716**. The guidewire **700** further comprises, instead of the proximal section **220** for example, two sections, an intermediate pump delivery section **730** and a backloading proximal section **740**. The backloading proximal section **740** is coupled to the intermediate pump delivery section **730** by a transition area **732**. A material of the backloading proximal section **740** is different than a material of the intermediate pump delivery section **730**. The transition area **732** can be a conical section with a first diameter equal to the diameter of the backloading proximal section **740** and a second diameter equal to the diameter of the intermediate pump delivery section. Alternatively, the transition **732** may be a weld. The transition **732** may be

an abrupt transition resulting from joining two sections of different flexibilities in a butt joint or a stepped joint. Alternatively, the transition **732** can be gradual by employing a composite construction such that the content of a more flexible material (e.g., the main material of the backloading proximal section **740**) gradually displaces a stiffer material (e.g., the main material of the intermediate pump delivery section **730**). The use of different materials for the backloading proximal section **740** and the intermediate pump delivery section **730** allows one section to be optimized for insertion (e.g., the backloading proximal section **740**) and another to be optimized for pump delivery (e.g., the intermediate pump delivery section **730**). This may also reduce the guidewire's weight, cost, or both.

(29) FIG. **8** shows a lateral cross-section of a fourth illustrative guidewire **800** according to certain embodiments. The guidewire **800** includes a soft distal section **810**, a distal tip **812**, a coil wire **814**, a core wire **816**, a transition area **818**, an intermediate pump delivery section **830**, a transition area **832**, a backloading proximal section **840**, and a proximal tip **842**. The guidewire distal section **810**, which is inserted first into the patient, extends between the transition area **818** and the distal tip **812**. The soft distal section **810** can include a core wire **816** with a diameter which decreases from the transition region **838** to the distal tip **812**. The soft distal section **810** can also include a coil wire **814** which is coiled around the core wire **816**. The guidewire **800** further comprises, instead of the proximal section **220** for example, two sections, an intermediate pump delivery section **830**, and a backloading proximal section **840** coupled to the intermediate pump delivery section **830** by a transition area **832**. The intermediate pump delivery section **830** can have a first structure which is a core wire with a constant diameter. The backloading proximal section **840** can have a second structure which includes a core wire **843**, and a coil wire **844**. The core wire **816** can have a constant diameter. Alternatively, the core wire **843** can be tapered to have a first larger diameter adjacent to the transition area **832**, and a second smaller diameter adjacent to the proximal tip **842**. The coil wire **844** may be wrapped around the core wire **843**. Similar to the core wire **843**, the coil wire **844** may have a constant diameter, or may have a diameter which decreases from transition area **832** to proximal tip **842**.

(30) The coil wire **844** can impart resistance to radial deformation and can allow the guidewire **800** to regain its original shape even after deformations that it may be subjected to during placement or manipulation in the heart. The coil wire **844** may have any number of cross-sectional shapes including but not limited to round or rectangular cross-sections. The coil wire **844** may also have varying axial density that varies the elasticity or flexibility of the backloading proximal section **840**.

(31) FIGS. **9** and **10** are tables summarizing the dimensions and material properties of exemplary embodiments of the guidewire shown in FIGS. **2** and **5-8**.

(32) FIG. **9** shows a table indicating diameter, length, function, materials and coating for a related art guidewire (e.g., guidewire **200** in FIG. **2**) and an improved guidewire with a proximal section and a distal section (e.g., guidewire in any of FIGS. **5-8**). FIG. **9** indicates that the outer diameter **D1** of a related art guidewire is 0.025" which is the same as the outer diameter **D1** of the improved guidewire distal section. FIG. **9** indicates that the diameter of the improved guidewire proximal section **D2** is 0.018". The function of the related art guidewire is both steering and guiding, whereas the improved guidewire proximal section is used for steering and the improved guidewire distal section is used for guiding. FIG. **9** includes an exemplary list of materials for both the related art guidewire and the improved guidewire (proximal section and distal section). For example, the related art guidewire is generally made of stainless steel. The improved guidewire proximal section may be made of a stainless core wire with a coating, coil jacket or plastics tubing jacket. Alternatively, the proximal section may be made of a Nitinol wire or plastics string. The proximal section of the improved guidewire may be coated. Similarly, the distal section of an improved guidewire may be made of a stainless core wire with a coating, coils jacket or plastics tube jacket. Alternatively, the improved guidewire distal section may be made of Nitinol wire.

(33) FIG. 10 shows force metrics for a related art guidewire proximal section (e.g., 220 in FIG. 2) with an outer diameter of 0.025" and for an improved guidewire proximal section with an outer diameter of 0.018". The maximum backloading force through the pump when using the related art guidewire is 1.5 Newtons. This maximum backloading force is significantly reduced to 0.3 N with the improved proximal guidewire. Furthermore, with the improved guidewire, a maximum sliding force for the guidewire is reduced from 1.9 N of initial resistance to a 0.7 N stable sliding force.

(34) FIG. 11 shows a system 1100, with a percutaneous pump already backloaded on a guidewire according to one of the illustrative embodiments of FIGS. 5-8. The system 1100 includes a guidewire 1102, a pigtail 1104 and a cannula 1108. In the example of FIG. 11, the system 1100 is curved to follow the shape of the pulmonary valve (not shown). As a physician backloads the pump and cannula 1108 onto the guidewire 1102, the guidewire 1102 passes through the pigtail 1104, through the cannula 1108, and in the gap located between the blades and the housing of the pump.

(35) FIG. 12 shows a method 1200 for inserting a percutaneous pump according to certain implementations. The method 1200 may be implemented to insert a percutaneous pump, e.g., pump 101, onto a guidewire including but not limited to the guidewires described in any of the aforementioned implementations. The method 1200 may be implemented by inserting a guidewire, previously positioned into the artery of a patient, into a cannula through a distal end of the cannula (step 1210), the distal end of the cannula supporting the percutaneous pump. The method 1200 further includes pushing the guidewire through the percutaneous pump (step 1220). The guidewire pushed through the percutaneous pump includes a proximal section with a first stiffness and a first diameter, and a distal section connected to the proximal section, the distal section with a second stiffness and a second diameter greater than the first diameter. The distal section of the guidewire is stiffer than the proximal section to insert the cannula in a desired location without displacing the guidewire.

(36) At least one benefit of the method 1200 is that it reduces the number of insertions of the guidewire into the patient, and minimizes the risk of damage to the patient's arterial system. The method 1200 uses backloading of the guidewire to reduce the number of insertions into the patient. Another benefit of the method 1200 is that it facilitates insertion of the guidewire into a pump, such as the pump 101 of the blood pump assembly 100. This reduces the risk of damage during insertion.

(37) In alternative embodiments, the percutaneous pump may be backloaded on the guidewire prior to the guidewire being positioned inside the artery of the patient. In alternative embodiments, the percutaneous pump may be backloaded on the guidewire before the percutaneous pump is coupled to the cannula.

(38) Variations and modifications will occur to those of skill in the art after reviewing this disclosure. For example, in some implementations, any of the alternative embodiments described in FIGS. 5-8 may be combined. For example, the coil structure of the proximal end of the guidewire in FIG. 7 may be combined with the different guidewire materials described with respect to FIGS. 5-6. In another example, the coil structure of the proximal end of the guidewire in FIG. 7 may be combined with a weld transition described in relation to FIG. 6. The disclosed features may be implemented, in any combination and subcombination (including multiple dependent combinations and subcombinations), with one or more other features described herein. The various features described or illustrated above, including any components thereof, may be combined or integrated in other systems. Moreover, certain features may be omitted or not implemented.

(39) It is important to note that the constructions and arrangements of apparatuses or the components thereof as shown in the various exemplary implementations are illustrative only. Although only a few implementations have been described in detail in this disclosure, those skilled in the art who review this disclosure will readily appreciate that many modifications are possible (e.g., variations in sizes, dimensions, structures, shapes and proportions of the various elements, values of parameters, mounting arrangements, use of materials, colors, orientations, etc.) without

materially departing from the novel teachings and advantages of the subject matter disclosed. For example, elements shown as integrally formed may be constructed of multiple parts or elements, the position of elements may be reversed or otherwise varied, and the nature or number of discrete elements or positions may be altered or varied. The order or sequence of any process or method steps may be varied or re-sequenced according to alternative implementations. Other substitutions, modifications, changes and omissions may also be made in the design, operating conditions and arrangement of the various exemplary implementations without departing from the scope of the present disclosure.

(40) While various inventive implementations have been described and illustrated herein, those of ordinary skill in the art will readily envision a variety of other mechanisms and/or structures for performing the function and/or obtaining the results and/or one or more of the advantages described herein, and each of such variations and/or modifications is deemed to be within the scope of the inventive implementations described herein. More generally, those skilled in the art will readily appreciate that, unless otherwise noted, any parameters, dimensions, materials, and configurations described herein are meant to be exemplary and that the actual parameters, dimensions, materials, and/or configurations will depend upon the specific application or applications for which the inventive teachings is/are used. Those skilled in the art will recognize, or be able to ascertain using no more than routine experimentation, many equivalents to the specific inventive implementations described herein. It is, therefore, to be understood that the foregoing implementations are presented by way of example only and that, within the scope of the appended claims and equivalents thereto, inventive implementations may be practiced otherwise than as specifically described and claimed. Inventive implementations of the present disclosure are directed to each individual feature, system, article, material, kit, and/or method described herein. In addition, any combination of two or more such features, systems, articles, materials, kits, and/or methods, if such features, systems, articles, materials, kits, and/or methods are not mutually inconsistent, is included within the inventive scope of the present disclosure.

(41) For the purpose of this disclosure, the term “coupled” means the joining of two members directly or indirectly to one another. Such joining may be stationary or moveable in nature. Such joining may be achieved with the two members or the two members and any additional intermediate members being integrally formed as a single unitary body with one another or within the two members of the two members and any additional intermediate members being attached to one another. Such joining may be permanent in nature or may be removable or releasable in nature.

(42) The indefinite articles “a” and “an,” as used herein in the specification and in the claims, unless clearly indicated to the contrary, should be understood to mean “at least one.” As used herein in the specification and in the claims, “or” should be understood to have the same meaning as “and/or” as defined above. For example, when separating items in a list, “or” or “and/or” shall be interpreted as being inclusive, i.e., the inclusion of at least one, but also including more than one, of a number or list of elements, and, optionally, additional unlisted items. Only terms clearly indicated to the contrary, such as “only one of” or “exactly one of” will refer to the inclusion of exactly one element of a number or list of elements. In general, the term “or” as used herein shall only be interpreted as indicating exclusive alternatives (i.e., “one or the other but not both”) when preceded by terms of exclusivity, such as “either,” “one of,” “only one of,” or “exactly one of.”

(43) In the claims, as well as in the specification above, all transitional phrases such as “comprising,” “including,” “carrying,” “having,” “containing,” “involving,” “holding,” “composed of,” and the like are to be understood to be open-ended, i.e., to mean including but not limited to.

(44) The claims should not be read as limited to the described order or elements unless stated to that effect. It should be understood that various changes in form and detail may be made by one of ordinary skill in the art without departing from the spirit and scope of the appended claims. All implementations that come within the spirit and scope of the following claims and equivalents thereto are claimed.

(45) Examples of changes, substitutions, and alterations are ascertainable by one skilled in the art and could be made without departing from the scope of the information disclosed herein. All references cited herein are incorporated by reference in their entirety and made part of this application.

## Claims

1. A method for inserting a pump configured for insertion into a vascular system, comprising: inserting a guidewire into a vasculature of a patient; inserting the guidewire into a cannula through a distal end of the cannula, the distal end of the cannula supporting a pump, wherein the pump comprises one or more blades; and pushing the guidewire through the pump, wherein inserting the guidewire into the vasculature of the patient occurs prior to inserting the guidewire into the cannula, and wherein the guidewire includes a proximal section with a first stiffness and a distal section with a second stiffness greater than the first stiffness, and wherein a third stiffness of the one or more blades of the pump is greater than the first stiffness of the guidewire.
2. The method of claim 1, wherein inserting the guidewire into the cannula comprises inserting the proximal section of the guidewire into the distal end of the cannula.
3. The method of claim 1, wherein the pump comprises a pump housing and one or more blades, and wherein pushing the guidewire through the pump comprises pushing the guidewire through a gap between the pump housing and the one or more blades.
4. The method of claim 1, wherein inserting the guidewire into the cannula comprises inserting the proximal section of the guidewire into an extension connected to the distal end of the cannula, followed by inserting the proximal section of the guidewire into the distal end of the cannula.
5. The method of claim 4, wherein the pump comprises a pump housing and one or more blades, and wherein pushing the guidewire through the pump comprises pushing the guidewire through a gap between the pump housing and the one or more blades.
6. The method of claim 1, wherein inserting the guidewire into the vasculature of the patient comprises inserting the guidewire into a femoral artery of the patient.
7. The method of claim 1, further comprising inserting the pump into a right ventricle of the patient.
8. The method of claim 1, further comprising inserting the pump into a left ventricle of the patient.
9. The method of claim 1, further comprising removing the guidewire from the vascular system of the patient.
10. A method for inserting a pump configured for insertion into a vascular system, comprising: inserting the guidewire into a cannula through a distal end of the cannula, the distal end of the cannula supporting a pump, wherein the pump comprises one or more blades; pushing the guidewire through the pump; and inserting the guidewire into a vasculature of a patient, wherein inserting the guidewire into the cannula occurs prior to inserting the guidewire into the vasculature of the patient, and wherein the guidewire includes a proximal section with a first stiffness and a distal section with a second stiffness greater than the first stiffness, and wherein a third stiffness of the one or more blades of the pump is greater than the first stiffness of the guidewire.
11. The method of claim 10, wherein inserting the guidewire into the cannula comprises inserting the proximal section of the guidewire into the distal end of the cannula.
12. The method of claim 10, wherein the pump comprises a pump housing and one or more blades, and wherein pushing the guidewire through the pump comprises pushing the guidewire through a gap between the pump housing and the one or more blades.
13. The method of claim 10, wherein inserting the guidewire into the cannula comprises inserting the proximal section of the guidewire into an extension connected to the distal end of the cannula, followed by inserting the proximal section of the guidewire into the distal end of the cannula.
14. The method of claim 13, wherein the pump comprises a pump housing and one or more blades,

and wherein pushing the guidewire through the pump comprises pushing the guidewire through a gap between the pump housing and the one or more blades.

15. The method of claim 10, wherein inserting the guidewire into the vasculature of the patient comprises inserting the guidewire into a femoral artery of the patient.

16. The method of claim 10, further comprising inserting the pump into a right ventricle of the patient.

17. The method of claim 10, further comprising inserting the pump into a left ventricle of the patient.

18. The method of claim 10, further comprising removing the guidewire from the vascular system of the patient.

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